

claude-windows / 188dd06f-9c99-487c-aa7d-b0bee83c629c

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\188dd06f-9c99-487c-aa7d-b0bee83c629c\subagents\workflows\wf\_416cc0e5-a35\agent-af9ccd3205ca9474b.jsonl`
- SHA-256: `bea3e035e66d5e54e22c667419f71e70452ffcb4e4e56924a5b9593143cd6052`
- Source modified: `2026-06-22T09:45:21+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf\_416cc0e5-a35`
- session_id: `188dd06f-9c99-487c-aa7d-b0bee83c629c`

Transcript

1. user (2026-06-22T09:42:12.646Z)

Review the NIGHTLY REEL-COUNT engine-regression change under C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/nightly-e2e (a NEW, additive PR). It runs the configured WASB pipeline on a GCP GPU VM nightly, asserts reel-count (==) + quality metrics + cost, emails a report, self-deletes. Files: scripts/nightly_reelcount.py, scripts/nightly_email.py, deploy/gcp/nightly_regression/run_on_vm.sh, deploy/gcp/nightly_regression/launch.sh, deploy/gcp/nightly_regression/register_scheduler.sh, deploy/gcp/nightly_regression/Dockerfile.launcher, eval_baselines/reelcount_manifest.json. Read them under that absolute path. Cite file:line. Report only REAL defects (correctness/safety/security that would bite in production or a regression that escapes), not style. Return raw structured findings.

LENS: PYTHON CORRECTNESS in scripts/nightly_reelcount.py + scripts/nightly_email.py. Scrutinize: compare() (reel-count EXACT, error?regression, quality tolerance, corpus-drift stale, regression-beats-stale), exit-code mapping, cost_report() math (per-second rate, billing usage/rates keys), summarize_results, the re-run-once mismatch guard (does it actually re-process + adopt the second value correctly?), the dotted config override (_set_dotted), exception handling (does any except mask a real regression as a pass?), --update-baseline path, the email graceful-skip. Flag any path where a true regression could exit 0, or a crash exits 0/masks the verdict.

2. assistant (2026-06-22T09:42:14.932Z)

I'll conduct a thorough security and correctness review of the nightly reel-count engine regression pipeline. Let me start by reading the key files.

3. user (2026-06-22T09:42:16.246Z)

```
1      """Nightly END-TO-END engine regression ? reel-count + quality + cost (the GCP-VM
2      nightly).
3      Runs the **configured + checkpointed** pipeline (the real WASB GPU detector, as a
4      deployment serves
5      it) end-to-end on 2-3 short owned golden videos and asserts the **number of reels does
6      not change**
7      vs a frozen baseline. Unlike ``scripts/nightly_regression.py`` (PR #202 ? a CPU re-score
8      of CACHED
9      trajectories, the windowing/scorer only), this one runs the *full* pipeline on the *real
10     model*, so it
11     needs a GPU ? hence it runs on an ephemeral GCP GPU VM
12     (`deploy/gcp/nightly_regression/`), emails the
13     owner the report, and self-deletes. It is **default-additive**: a tripwire, not the
14     promotion gate.
15     What it asserts / reports, per the owner's ask ("quality metrics + regression report +
```

```

cost"):
11     ? REEL COUNT per video ? ``len(play_segments)`` after segmentation; the detected
highlight clips.
12     Pure-CV / WASB detection is deterministic for a fixed (video, config, weights), so
this is asserted
13     **EXACT** (with a re-run-once guard for the one known flake: a transient DB-insert-
None, see $guard).
14     ? QUALITY METRICS per video (when golden labels are provided) ? the R2 tolerance-match
block from
15     ``backend.eval.rally_seg_eval`` (tol_f1 / f1@0.5 / f1@0.25), compared with a small
tolerance.
16     ? COST of the run ? VM uptime (or processing wall) × machine $/hr via
``backend.billing`` ? USD + INR.
17
18     Extensible pattern (add a video = one manifest row + re-freeze):
19     ``eval_baselines/reelcount_manifest.json`` ? the videos (gs:// / local / gdrive://
URIs) + segmenter
20     ``eval_baselines/reelcount_baseline.json`` ? the frozen expected counts (regenerated,
never hand-edited)
21
22     Usage
23     -----
24     Freeze the baseline from the current configured+checkpointed engine (after an intended
change, or first run):
25     python scripts/nightly_reelcount.py --update-baseline
26
27     Nightly check (exit 1 on regression; writes output/nightly_reelcount/<date>.md; emails
if --email):
28     python scripts/nightly_reelcount.py --email
29
30     Exit codes: 0 pass · 1 regression (reel-count changed / a video errored / quality
dropped) · 2 operational
31     (manifest/videos missing, nothing ran) · 3 no baseline · 4 stale (corpus/segmenter
changed ? re-freeze).
32     """
33     from __future__ import annotations
34
35     import argparse
36     import datetime as _dt
37     import json
38     import os
39     import sys
40     from dataclasses import dataclass, field
41     from typing import Any, Dict, List, Optional, Tuple
42
43     REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
44     if REPO_ROOT not in sys.path:
45         sys.path.insert(0, REPO_ROOT)
46
47     # billing is pure (typing only) ? safe to import at module top.
48     from backend import billing # noqa: E402
49
50     DEFAULT_MANIFEST = os.path.join(REPO_ROOT, "eval_baselines", "reelcount_manifest.json")
51     DEFAULT_BASELINE = os.path.join(REPO_ROOT, "eval_baselines", "reelcount_baseline.json")
52     DEFAULT_REPORT_DIR = os.path.join(REPO_ROOT, "output", "nightly_reelcount")
53
54     #: Quality metrics we report + (when labels exist) gate ? higher is better, small
tolerance.
55     QUALITY_HIGHER: Tuple[str, ...] = ("tol_f1", "f1@0.5", "f1@0.25")
56     DEFAULT_QUALITY_TOL = 0.02 # quality is float/labeled ? a looser tripwire than the
exact reel-count
57     DEFAULT_FX_INR_PER_USD = 83.0
58     #: Fallback machine price if the manifest doesn't pin one (T4 on n1-standard-8, asia-
south1, ~mid-2026).
59     DEFAULT_MACHINE_USD_PER_HR = 0.35

```

```

60
61
62 @dataclass(frozen=True)
63 class Tolerances:
64     quality_tol: float = DEFAULT_QUALITY_TOL
65
66     def to_dict(self) -> Dict[str, Any]:
67         return {"quality_tol": self.quality_tol}
68
69
70 # ----- #
71 # Pure cost math (wraps backend.billing ? no IO).
72 # ----- #
73 def cost_report(billed_seconds: float, machine_usd_per_hr: float,
74                 fx_inr_per_usd: float, gpu_seconds: float = 0.0) -> Dict[str, Any]:
75     """Run cost from billed wall-seconds × machine rate ? USD + INR (``backend.billing``
76 arithmetic).
77     ``billed_seconds`` is the VM uptime (boot?delete) when the orchestrator passes
78 ``--vm-uptime-seconds``;
79     otherwise it's the in-process pipeline wall time (a lower bound ? the VM also pays
80 for boot/install)."""
81     rate_per_sec = float(machine_usd_per_hr) / 3600.0
82     usage = {billing.WALL_SECONDS: float(billed_seconds)}
83     rates = {billing.WALL_SECONDS: rate_per_sec}
84     usd, breakdown = billing.compute_cost_usd(usage, rates)
85     return {
86         "billed_seconds": round(float(billed_seconds), 1),
87         "gpu_seconds": round(float(gpu_seconds), 1),
88         "machine_usd_per_hr": float(machine_usd_per_hr),
89         "usd": usd,
90         "inr": billing.to_inr(usd, fx_inr_per_usd),
91         "breakdown": breakdown,
92     }
93
94 # ----- #
95 # Pure summary + comparison core (no IO ? unit-testable with plain dicts).
96 # ----- #
97 def summarize_results(run_results: List[Dict[str, Any]], segmenter: str,
98                       cost: Dict[str, Any]) -> Dict[str, Any]:
99     """Compact, comparable snapshot from per-video run results.
100     Each run result: ``{name, reel_count, ok, error, quality: {tol_f1,...}|None,
101 wall_seconds}``.
102     A video that errored keeps ``reel_count=None`` + ``ok=False`` (so the diff flags it,
103 never hides it)."""
104     per_video: Dict[str, Any] = {}
105     for r in run_results:
106         per_video[r["name"]] = {
107             "reel_count": r.get("reel_count"),
108             "ok": bool(r.get("ok", False)),
109             "error": r.get("error"),
110             "quality": r.get("quality"),
111         }
112     return {"segmenter": segmenter, "cost": cost, "per_video": per_video,
113           "n": len(per_video)}
114
115 @dataclass
116 class Finding:
117     video: str
118     metric: str
119     corpus: str
120     kind: str
121     # "reel_count" | "error" | a quality key | "segmenter" |
122     # "regression" | "improvement" | "stale"

```

```

119     baseline: Optional[float] = None
120     current: Optional[float] = None
121     detail: str = ""
122
123     def message(self) -> str:
124         if self.kind == "stale":
125             return f"[STALE] {self.metric}: {self.detail}"
126         if self.metric == "error":
127             return f"[REGRESSION] {self.video}: pipeline ERROR ? {self.detail}"
128         b = "?" if self.baseline is None else f"{self.baseline:g}"
129         c = "?" if self.current is None else f"{self.current:g}"
130         tag = "REGRESSION" if self.kind == "regression" else "improvement"
131         return f"[{tag}] {self.video}/{self.metric}: {b} ? {c}"
132
133
134 @dataclass
135 class NightlyResult:
136     findings: List[Finding] = field(default_factory=list)
137     added: List[str] = field(default_factory=list)
138     removed: List[str] = field(default_factory=list)
139
140     @property
141     def regressions(self) -> List[Finding]:
142         return [f for f in self.findings if f.kind == "regression"]
143
144     @property
145     def improvements(self) -> List[Finding]:
146         return [f for f in self.findings if f.kind == "improvement"]
147
148     @property
149     def stale(self) -> List[Finding]:
150         return [f for f in self.findings if f.kind == "stale"]
151
152     def overall_status(self) -> str:
153         if self.regressions:
154             return "REGRESSION"
155         if self.stale:
156             return "STALE"
157         return "PASS"
158
159     def exit_code(self) -> int:
160         if self.regressions:
161             return 1
162         if self.stale:
163             return 4
164         return 0
165
166
167     def compare(baseline: Dict[str, Any], current: Dict[str, Any], tols: Tolerances) ->
NightlyResult:
168         """Diff current run vs frozen baseline.
169
170         * Segmenter changed ? STALE (numbers not comparable; re-freeze).
171         * Corpus (video set) changed ? STALE + report added/removed; still diff the
INTERSECTION.
172         * Per shared video: reel_count EXACT (any change = regression); a pipeline ERROR =
regression;
173         quality metrics drop beyond ``quality_tol`` = regression (rise = improvement).
174         """
175         res = NightlyResult()
176         if baseline.get("segmenter") != current.get("segmenter"):
177             res.findings.append(Finding("", "segmenter", "stale",
178                                     detail=f"{baseline.get('segmenter')} ?
{current.get('segmenter')}"))
179         return res

```

```

180
181     b_pv, c_pv = baseline.get("per_video", {}), current.get("per_video", {})
182     res.added = sorted(set(c_pv) - set(b_pv))
183     res.removed = sorted(set(b_pv) - set(c_pv))
184     if res.added or res.removed:
185         res.findings.append(Finding("", "corpus", "stale",
186                                   detail=f"added={res.added} removed={res.removed}"))
187
188     for v in sorted(set(b_pv) & set(c_pv)):
189         b, c = b_pv[v], c_pv[v]
190         # 1) pipeline must still succeed
191         if not c.get("ok", False) or c.get("reel_count") is None:
192             res.findings.append(Finding(v, "error", "regression",
193                                       detail=str(c.get("error") or "no reel_count
194 produced"))))
195             continue
196         # 2) reel count ? EXACT
197         bc, cc = b.get("reel_count"), c.get("reel_count")
198         if bc is not None and cc is not None and cc != bc:
199             res.findings.append(Finding(v, "reel_count", "regression", float(bc),
200 float(cc)))
201         # 3) quality metrics ? tolerance (only if both sides have them)
202         bq, cq = b.get("quality") or {}, c.get("quality") or {}
203         for m in QUALITY_HIGHER:
204             bv, cv = bq.get(m), cq.get(m)
205             if bv is None or cv is None:
206                 continue
207             if cv < bv - tols.quality_tol:
208                 res.findings.append(Finding(v, m, "regression", bv, cv))
209             elif cv > bv + tols.quality_tol:
210                 res.findings.append(Finding(v, m, "improvement", bv, cv))
211
212     return res
213
214 # ----- #
215 # Report formatting (pure).
216 # ----- #
217 def _q(qd: Optional[Dict[str, Any]], key: str) -> str:
218     if not qd or qd.get(key) is None:
219         return "?"
220     return f"{qd[key]:.3f}"
221
222 def format_report(baseline: Dict[str, Any], current: Dict[str, Any],
223                  result: NightlyResult, date: str, head: str) -> str:
224     status = result.overall_status()
225     badge = {"PASS": "? PASS", "REGRESSION": "? REGRESSION",
226             "STALE": "? STALE (re-freeze the baseline)"}[status]
227     cost = current.get("cost", {})
228     L: List[str] = [
229         f"# Nightly engine regression (reel-count + quality + cost) ? {date}",
230         "",
231         f"**Status: {badge}** · exit code {result.exit_code()} · segmenter
232 {current.get('segmenter')}",
233         "",
234         f"HEAD {head} · baseline {baseline.get('git_commit', '?')} (frozen
235 {baseline.get('generated_at', '?')})",
236         f"- **Run cost: ${cost.get('usd', 0):.4f} (?{cost.get('inr', 0):.2f})** · "
237         f"billed {cost.get('billed_seconds', 0):.0f}s @ ${cost.get('machine_usd_per_hr',
238 0)}/hr",
239         "",
240     ]
241     if result.regressions:
242         L += ["## ? Regressions", ""] + [f"- {f.message()}" for f in result.regressions]
243     + [""]

```

```

239         if result.stale:
240             L += ["## ? Stale / not comparable", ""] + [f"- {f.message()}" for f in
result.stale]
241             L += ["- Action: `python scripts/nightly_reelcount.py --update-baseline` (after
confirming the change is intended).", ""]
242
243         # Per-video table.
244         b_pv, c_pv = baseline.get("per_video", {}), current.get("per_video", {})
245         L += ["## Per-video", "",
246             "| video | reels (base?now) | tol_f1 (base?now) | f1@0.5 | status |",
247             "|---|---|---|---|---|"]
248         for v in sorted(set(b_pv) | set(c_pv)):
249             b, c = b_pv.get(v, {}), c_pv.get(v, {})
250             bc = b.get("reel_count")
251             cc = c.get("reel_count")
252             rc = f"'?' if bc is None else bc"?'ERROR' if not c.get('ok', True) else ('?'
if cc is None else cc)})"
253             bq, cq = b.get("quality"), c.get("quality")
254             tf = f"_q(bq, 'tol_f1')?_q(cq, 'tol_f1')}"
255             f5 = f"_q(bq, 'f1@0.5')?_q(cq, 'f1@0.5')}"
256             vstatus = "?" if any(f.video == v and f.kind == "regression" for f in
result.findings) else "?"
257             L.append(f"| {v} | {rc} | {tf} | {f5} | {vstatus} |")
258         L.append("")
259
260         if result.improvements:
261             L += ["_Improvements over baseline (re-freeze to ratchet up):_"] \
262                 + [f"- {f.message()}" for f in result.improvements] + [""]
263         L += ["---",
264             "_Full configured+checkpointed pipeline on the real model (GCP GPU VM).
Tripwire only ? does "
265             "not change the promotion gate. Complements PR #202's cached-trajectory CPU
re-score._"]
266         return "\n".join(L)
267
268
269         # ----- #
270         # IO shell ? running the configured pipeline per video (needs backend + GPU + video).
271         # ----- #
272         def _load_gts(labels_ref: Optional[str]) -> Optional[List[Tuple[float, float]]]:
273             """Load ground-truth rally intervals from a golden-features JSON (`gts` key) or a
rallies CSV
274             (`start,end` columns / pairs). Returns None when no labels are configured."""
275             if not labels_ref:
276                 return None
277             from backend.storage import fetch
278             local = fetch(labels_ref)
279             if local.endswith(".json"):
280                 with open(local, encoding="utf-8") as fh:
281                     data = json.load(fh)
282                     return [(float(s), float(e)) for s, e in (data.get("gts") or [])]
283             # CSV: header start,end (or first two numeric columns)
284             import csv
285             out: List[Tuple[float, float]] = []
286             with open(local, encoding="utf-8") as fh:
287                 for row in csv.reader(fh):
288                     try:
289                         out.append((float(row[0]), float(row[1])))
290                     except (ValueError, IndexError):
291                         continue
292             return out
293
294
295         def _segments_to_intervals(segs: List[Dict[str, Any]]) -> List[Tuple[float, float]]:
296             """(start, end)` pairs from segmenter result dicts, tolerant of the two key

```

```

conventions in the
297     codebase (`start_time`/`end_time` ? motion/DB shape ? or `start`/`end`).
Used only for the
298     optional quality metrics; the reel COUNT is `len(segs)` regardless, so a key
mismatch degrades
299     quality scoring gracefully (skipped) without ever miscounting reels."""
300     out: List[Tuple[float, float]] = []
301     for s in segs:
302         start = s.get("start_time", s.get("start"))
303         end = s.get("end_time", s.get("end"))
304         if start is not None and end is not None:
305             out.append((float(start), float(end)))
306     return out
307
308
309     def run_one_video(video: Dict[str, Any], segmenter: str, config: Any,
310                      rerun_on_mismatch: bool = True,
311                      baseline_count: Optional[int] = None) -> Dict[str, Any]:
312         """Run the configured pipeline on one video ? reel count (+ quality vs labels). Lazy
backend imports.
313
314         The re-run-once guard (the one known non-determinism ? a transient
`add_play_segment`?None drop,
315         database.py): if the count != the frozen baseline, re-process the video ONCE; a
transient drop won't
316         reproduce, a real change will."""
317         import random
318         import time
319
320         from backend.eval import rally_seg_eval
321         from backend.infrastructure.database import Database
322         from backend.pipeline.segmenters import segmenter_registry
323         from backend.results import Failure
324         from backend.storage import fetch
325         from backend.utils.hashing import compute_video_id
326
327         name = video["name"]
328         try:
329             local = fetch(video["uri"])
330         except Exception as e: # noqa: BLE001 ? surface, never hide
331             return {"name": name, "reel_count": None, "ok": False, "error": f"fetch failed:
{e}",
332                    "quality": None, "wall_seconds": 0.0}
333
334         gts = _load_gts(video.get("labels_uri"))
335         seg_cls = segmenter_registry.get(segmenter)
336         if not seg_cls:
337             return {"name": name, "reel_count": None, "ok": False,
338                    "error": f"segmenter '{segmenter}' not registered", "quality": None,
"wall_seconds": 0.0}
339
340         def _process(db_path: str) -> Tuple[Optional[int], Optional[List[Tuple[float,
float]]], float, Optional[str]]:
341             random.seed(0) # close motion.py's unseeded metadata RNG so secondary fields
are reproducible too
342             db = Database(db_path)
343             video_id = compute_video_id(local)
344             t0 = time.monotonic()
345             result = seg_cls(db, config).process_video(video_id, local)
346             wall = time.monotonic() - t0
347             if isinstance(result, Failure):
348                 return None, None, wall, getattr(result, "message", "segmenter failed")
349             segs = result.value if hasattr(result, "value") else result
350             intervals = _segments_to_intervals(segs)
351             return len(segs), intervals, wall, None

```

```

352
353 import tempfile
354 total_wall = 0.0
355 with tempfile.TemporaryDirectory(prefix="nightly_reel_") as td:
356     count, intervals, wall, err = _process(os.path.join(td, "run1.db"))
357     total_wall += wall
358     if err is not None:
359         return {"name": name, "reel_count": None, "ok": False, "error": err,
360                "quality": None, "wall_seconds": round(total_wall, 2)}
361     # re-run-once guard for the transient DB-drop flake
362     if rerun_on_mismatch and baseline_count is not None and count != baseline_count:
363         count2, intervals2, wall2, err2 = _process(os.path.join(td, "run2.db"))
364         total_wall += wall2
365         if err2 is None and count2 is not None:
366             count, intervals = count2, intervals2 # the reproduced (or corrected)
value
367
368     quality = None
369     if gts and intervals is not None:
370         row = rally_seg_eval.score_intervals(intervals, gts)
371         quality = {k: row[k] for k in ("tol_f1", "f1@0.5", "f1@0.25",
"tol_precision", "tol_recall")}
372         if k in row}
373
374     return {"name": name, "reel_count": count, "ok": True, "error": None,
375            "quality": quality, "wall_seconds": round(total_wall, 2)}
376
377
378 def load_manifest(path: str) -> Dict[str, Any]:
379     with open(path, encoding="utf-8") as fh:
380         return json.load(fh)
381
382
383 def run_corpus(manifest: Dict[str, Any], baseline: Optional[Dict[str, Any]],
384               rerun_on_mismatch: bool = True) -> Tuple[List[Dict[str, Any]], float]:
385     """Run every manifest video through the configured pipeline. Returns (run_results,
total_wall_s)."""
386     from backend.config import load_config
387
388     config = load_config()
389     # apply manifest config overrides onto the typed config (dotted keys)
390     for dotted, val in (manifest.get("config_overrides") or {}).items():
391         _set_dotted(config, dotted, val)
392     segmenter = manifest.get("segmenter") or getattr(getattr(config, "indexing", None),
"default_segmenter", "motion")
393     b_pv = (baseline or {}).get("per_video", {})
394     results: List[Dict[str, Any]] = []
395     total_wall = 0.0
396     for v in manifest.get("videos", []):
397         base_count = (b_pv.get(v["name"]) or {}).get("reel_count")
398         r = run_one_video(v, segmenter, config, rerun_on_mismatch, base_count)
399         total_wall += float(r.get("wall_seconds") or 0.0)
400         results.append(r)
401     return results, total_wall
402
403
404 def _set_dotted(config: Any, dotted: str, value: Any) -> None:
405     """Best-effort dotted-path set onto a pydantic config (e.g.
'indexing.motion_threshold')."""
406     parts = dotted.split(".")
407     obj = config
408     for p in parts[:-1]:
409         obj = getattr(obj, p, None)
410         if obj is None:
411             return

```

```

412         try:
413             setattr(obj, parts[-1], value)
414         except Exception: # noqa: BLE001 ? overrides are best-effort; a bad key shouldn't
crash the run
415             pass
416
417
418     def _git_head() -> str:
419         import subprocess
420         try:
421             out = subprocess.run(["git", "-C", REPO_ROOT, "rev-parse", "--short", "HEAD"],
422                                 capture_output=True, text=True, timeout=10)
423             return out.stdout.strip() or "?"
424         except Exception: # noqa: BLE001
425             return "?"
426
427
428     def load_baseline(path: str) -> Optional[Dict[str, Any]]:
429         if not os.path.exists(path):
430             return None
431         with open(path, encoding="utf-8") as fh:
432             return json.load(fh)
433
434
435     def save_baseline(path: str, snapshot: Dict[str, Any]) -> None:
436         out = dict(snapshot)
437         out["generated_at"] = _dt.date.today().isoformat()
438         out["git_commit"] = _git_head()
439         out["_doc"] = ("Frozen nightly reel-count baseline. Regenerate with `python
scripts/nightly_reelcount.py "
440                       "--update-baseline` after an INTENDED engine change or when adding a
video. Tied to the "
441                       "configured+checkpointed engine + the manifest corpus.")
442         out.pop("cost", None) # cost is per-run, not part of the frozen baseline
443         os.makedirs(os.path.dirname(path), exist_ok=True)
444         tmp = path + ".tmp"
445         with open(tmp, "w", encoding="utf-8") as fh:
446             json.dump(out, fh, indent=2, sort_keys=True)
447             fh.write("\n")
448         os.replace(tmp, path)
449
450
451     def build_parser() -> argparse.ArgumentParser:
452         p = argparse.ArgumentParser(description="Nightly E2E engine regression: reel-count +
quality + cost.")
453         p.add_argument("--manifest", default=DEFAULT_MANIFEST)
454         p.add_argument("--baseline", default=DEFAULT_BASELINE)
455         p.add_argument("--report-dir", default=DEFAULT_REPORT_DIR)
456         p.add_argument("--quality-tol", type=float, default=DEFAULT_QUALITY_TOL)
457         p.add_argument("--vm-uptime-seconds", type=float, default=None,
458                       help="authoritative VM uptime (boot?delete) for the cost line; "
459                       "defaults to the in-process pipeline wall time (a lower bound)")
460         p.add_argument("--no-rerun-guard", action="store_true",
461                       help="disable the re-run-once-on-mismatch guard (the DB-drop flake
mitigation)")
462         p.add_argument("--update-baseline", action="store_true",
463                       help="freeze the current reel counts + quality as the new baseline
instead of checking")
464         p.add_argument("--email", action="store_true", help="email the report (see
scripts/nightly_email.py)")
465         p.add_argument("--email-to", default=None, help="override the report recipient")
466         p.add_argument("--date", default=None, help="override the report date (YYYY-MM-DD;
for tests)")
467         p.add_argument("--no-report", action="store_true", help="skip writing the dated
report file")

```

```

468         return p
469
470
471     def main(argv: Optional[List[str]] = None) -> int:
472         import logging
473         logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")
474         log = logging.getLogger("nightly_reelcount")
475         args = build_parser().parse_args(argv)
476         tols = Tolerances(quality_tol=args.quality_tol)
477         date = args.date or _dt.date.today().isoformat()
478
479         if not os.path.exists(args.manifest):
480             log.error("manifest not found: %s", args.manifest)
481             return 2
482         manifest = load_manifest(args.manifest)
483         baseline = load_baseline(args.baseline)
484
485         try:
486             run_results, total_wall = run_corpus(manifest, baseline, rerun_on_mismatch=not
args.no_rerun_guard)
487         except Exception as e: # noqa: BLE001 ? operational failure (deps/config) ? exit 2,
never a false "pass"
488             log.error("run failed: %s", e)
489             return 2
490         if not run_results:
491             log.error("no videos in manifest %s", args.manifest)
492             return 2
493
494         billed = args.vm_uptime_seconds if args.vm_uptime_seconds is not None else
total_wall
495         cost = cost_report(
496             billed_seconds=billed,
497             machine_usd_per_hr=float(manifest.get("machine_usd_per_hr",
DEFAULT_MACHINE_USD_PER_HR)),
498             fx_inr_per_usd=float(manifest.get("fx_inr_per_usd", DEFAULT_FX_INR_PER_USD)),
499             gpu_seconds=total_wall,
500         )
501         segmenter = manifest.get("segmenter", "motion")
502         snapshot = summarize_results(run_results, segmenter, cost)
503         snapshot["git_commit"] = _git_head()
504         snapshot["generated_at"] = date
505
506         if args.update_baseline:
507             save_baseline(args.baseline, snapshot)
508             log.info("baseline frozen ? %s", args.baseline)
509             for r in run_results:
510                 log.info(" %s: reels=%s ok=%s", r["name"], r.get("reel_count"),
r.get("ok"))
511             return 0
512
513         if baseline is None:
514             log.error("no baseline at %s ? freeze one with --update-baseline (on the
configured engine).", args.baseline)
515             return 3
516
517         result = compare(baseline, snapshot, tols)
518         report = format_report(baseline, snapshot, result, date, snapshot["git_commit"])
519
520         if not args.no_report:
521             os.makedirs(args.report_dir, exist_ok=True)
522             md_path = os.path.join(args.report_dir, f"{date}.md")
523             with open(md_path, "w", encoding="utf-8") as fh:
524                 fh.write(report + "\n")
525             with open(os.path.join(args.report_dir, f"{date}.json"), "w", encoding="utf-8")
as fh:

```

```

526         json.dump({"date": date, "status": result.overall_status(), "exit_code":
result.exit_code(),
527                     "snapshot": snapshot, "regressions": [f.message() for f in
result.regressions]},
528                     fh, indent=2, sort_keys=True)
529         log.info("report ? %s", md_path)
530
531     print(report)
532
533     if args.email:
534         try:
535             from scripts.nightly_email import send_report
536             subject = f"[KhelSutra nightly] {result.overall_status()} ? reel-count
engine regression {date}"
537             sent = send_report(subject, report, to=args.email_to)
538             log.info("email %s", "sent" if sent else "skipped (no creds ? report still
written)")
539         except Exception as e: # noqa: BLE001 ? a broken mailer must NOT mask the
regression verdict
540             log.error("email failed (report still written): %s", e)
541
542         code = result.exit_code()
543         if code == 1:
544             log.error("REEL-COUNT REGRESSION (%d) ? see %s", len(result.regressions),
args.report_dir)
545         elif code == 4:
546             log.warning("baseline STALE ? corpus/segmenter changed; re-freeze with --update-
baseline")
547         else:
548             log.info("nightly reel-count check PASSED (cost $%.4f / ?%.2f)", cost["usd"],
cost["inr"])
549         return code
550
551
552     if __name__ == "__main__":
553         raise SystemExit(main())
554

```

4. user (2026-06-22T09:42:16.247Z)

```

1     """Email the nightly regression report ? stdlib SMTP, creds from env or GCP Secret
Manager.
2
3     Used by ``scripts/nightly_reelcount.py --email`` (and reusable by any nightly).
Deliberately
4     dependency-light (stdlib ``smtplib`` + ``email.message``) and **graceful**: if no SMTP
password can
5     be resolved it logs + returns ``False`` (the report is still written/printed ? a broken
mailer must
6     never mask a regression).
7
8     Credential resolution (first hit wins), so it works on a GCP VM, locally, or in tests:
9     ? Recipient: ``--email-to`` arg ? ``NIGHTLY_EMAIL_TO`` ? ``avi.dullu@gmail.com``
10    ? SMTP host: ``NIGHTLY_SMTP_HOST`` (default ``smtp.gmail.com``) . port
``NIGHTLY_SMTP_PORT`` (587, STARTTLS)
11    ? SMTP user: ``NIGHTLY_SMTP_USER`` (default = recipient)
12    ? SMTP pass: ``NIGHTLY_SMTP_PASS`` ? Secret Manager ``NIGHTLY_SMTP_SECRET``
13    (``projects/<p>/secrets/<name>/versions/latest``; via the google-cloud-
secret-manager
14    lib if installed, else the ``gcloud`` CLI). A Gmail **app password** is
the simplest.
15
16    Set the secret once (owner): echo -n '<app-password>' | gcloud secrets create nightly-
smtp-pass --data-file=-
17    and on the VM: export NIGHTLY_SMTP_SECRET=projects/<proj>/secrets/nightly-smtp-

```

```

pass/versions/latest
18             export NIGHTLY_SMTP_USER=<your-gmail>
NIGHTLY_EMAIL_TO=avi.dullu@gmail.com
19         """
20         from __future__ import annotations
21
22         import logging
23         import os
24         from email.message import EmailMessage
25         from typing import Optional
26
27         log = logging.getLogger("nightly_email")
28
29         DEFAULT_TO = "avi.dullu@gmail.com"
30         DEFAULT_HOST = "smtp.gmail.com"
31         DEFAULT_PORT = 587
32
33
34         def build_message(subject: str, body_md: str, from_addr: str, to_addr: str) ->
EmailMessage:
35             """A text/plain email carrying the markdown report (renders fine in any client).
Pure ? testable."""
36             msg = EmailMessage()
37             msg["Subject"] = subject
38             msg["From"] = from_addr
39             msg["To"] = to_addr
40             msg.set_content(body_md)
41             return msg
42
43
44         def resolve_secret(env: Optional[dict] = None) -> Optional[str]:
45             """Resolve the SMTP password from env or GCP Secret Manager. Returns None if
unresolvable."""
46             env = env if env is not None else os.environ
47             direct = env.get("NIGHTLY_SMTP_PASS")
48             if direct:
49                 return direct
50             secret_ref = env.get("NIGHTLY_SMTP_SECRET")
51             if not secret_ref:
52                 return None
53             # Prefer the client lib; fall back to the gcloud CLI (no extra dependency required).
54             try:
55                 from google.cloud import secretmanager # type: ignore
56                 client = secretmanager.SecretManagerServiceClient()
57                 return
client.access_secret_version(name=secret_ref).payload.data.decode("utf-8")
58             except Exception: # noqa: BLE001 ? lib absent or auth issue ? try the CLI
59                 pass
60             try:
61                 import subprocess
62                 name = secret_ref.split("/secrets/", 1)[1].split("/", 1)[0] if "/secrets/" in
secret_ref else secret_ref
63                 out = subprocess.run(["gcloud", "secrets", "versions", "access", "latest", "--
secret", name],
64                                     capture_output=True, text=True, timeout=30)
65                 return out.stdout.strip() if out.returncode == 0 and out.stdout.strip() else
None
66             except Exception as e: # noqa: BLE001
67                 log.warning("secret-manager resolution failed: %s", e)
68                 return None
69
70
71         def send_report(subject: str, body_md: str, to: Optional[str] = None,
72                         env: Optional[dict] = None) -> bool:
73             """Send the report. Returns True if sent, False if skipped (no creds) ? never raises

```

```

on missing creds.""
74     env = env if env is not None else os.environ
75     to_addr = to or env.get("NIGHTLY_EMAIL_TO") or DEFAULT_TO
76     host = env.get("NIGHTLY_SMTP_HOST", DEFAULT_HOST)
77     port = int(env.get("NIGHTLY_SMTP_PORT", DEFAULT_PORT))
78     user = env.get("NIGHTLY_SMTP_USER") or to_addr
79     password = resolve_secret(env)
80     if not password:
81         log.warning("no SMTP password (set NIGHTLY_SMTP_PASS or NIGHTLY_SMTP_SECRET) ?
email skipped.")
82         return False
83
84     import smtplib
85     msg = build_message(subject, body_md, from_addr=user, to_addr=to_addr)
86     with smtplib.SMTP(host, port, timeout=60) as s:
87         s.starttls()
88         s.login(user, password)
89         s.send_message(msg)
90     log.info("emailed report to %s", to_addr)
91     return True
92

```

5. user (2026-06-22T09:42:17.222Z)

```

1     #!/usr/bin/env bash
2     # Startup-script for the NIGHTLY engine-regression VM. Runs UNATTENDED on boot, then
DELETES the VM
3     # (cost rule: a running GPU VM is the only real cost). Survives SSH drops (it's a
startup-script, not
4     # an interactive session). Reads its config from instance metadata (set by launch.sh).
5     #
6     # Flow: Ops Agent ? pull repo @ HEAD from GCS ? build venv + WASB env ? stage
weights ?
7     #         run scripts/nightly_reelcount.py on the gs:// golden clips (emails the report)
?
8     #         upload report to GCS ? SELF-DELETE (even on failure, via the EXIT trap).
9     #
10    # Metadata keys (launch.sh sets these): repo-tgz-uri, git-sha, gcs-bucket, report-
prefix,
11    # weights-uri, email (true/false). NOT verified in CI ? dry-run via launch.sh on a
real VM first.
12    set -uo pipefail
13    exec > >(tee -a /var/log/nightly-regression.log) 2>&1
14    echo "==== nightly-regression startup $(date -u +%FT%TZ) ====="
15    BOOT_EPOCH=$(date +%s)
16
17    md() { curl -s -H "Metadata-Flavor: Google"
"http://metadata.google.internal/computeMetadata/v1/$1"; }
18    NAME=$(md instance/name)
19    ZONE=$(md instance/zone | awk -F/ '{print $NF}')
20    REPO_TGZ=$(md instance/attributes/repo-tgz-uri || true)
21    GIT_SHA=$(md instance/attributes/git-sha || echo unknown)
22    BUCKET=$(md instance/attributes/gcs-bucket || true)
23    REPORT_PREFIX=$(md instance/attributes/report-prefix || echo nightly/reports)
24    WEIGHTS_URI=$(md instance/attributes/weights-uri || true)
25    DO_EMAIL=$(md instance/attributes/email || echo true)
26    SMTP_SECRET=$(md instance/attributes/smtp-secret || true)
27    SMTP_USER=$(md instance/attributes/smtp-user || true)
28    EMAIL_TO=$(md instance/attributes/email-to || echo avi.dullu@gmail.com)
29    DATE=$(date -u +%F)
30
31    # ALWAYS delete the VM on exit (success OR failure) ? never leave a billing GPU VM
behind.
32    cleanup() {
33        echo "==== self-delete $NAME ($ZONE) ====="

```

```

34     gcloud compute instances delete "$NAME" --zone="$ZONE" --quiet || \
35     echo "WARN: self-delete failed ? DELETE MANUALLY: gcloud compute instances delete
$NAME --zone=$ZONE -q"
36 }
37 trap cleanup EXIT
38
39 fail_email() { # best-effort failure alert (the report email won't have fired on an
early crash)
40     echo "RUN FAILED ? attempting a failure alert"
41     if [ "$DO_EMAIL" = "true" ] && [ -d ~/rally ]; then
42         cd ~/rally 2>/dev/null && . .venv/bin/activate 2>/dev/null && \
43         NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
44         python - <<PY || true
45     from scripts.nightly_email import send_report
46     send_report("[KhelSutra nightly] ERROR ? engine regression VM run failed ($DATE)",
47         "The nightly engine-regression run failed on the VM before producing a
report.\n"
48         "See gs://$BUCKET/$REPORT_PREFIX/$DATE/ and the VM serial
log.\nGIT_SHA=$GIT_SHA", to="$EMAIL_TO")
49     PY
50     fi
51 }
52 trap 'fail_email' ERR
53
54 # 1) Observability (Ops Agent) ? same as create_gpu_vm.sh bakes in.
55 curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh && \
56     sudo bash add-google-cloud-ops-agent-repo.sh --also-install || echo "WARN: ops-agent
install failed (non-fatal)"
57
58 # 2) Pull the repo @ HEAD (staged to GCS by launch.sh ? no GitHub auth needed; uses the
worker SA).
59 mkdir -p ~/rally && cd ~/rally
60 gcloud storage cp "$REPO_TGZ" /tmp/rally.tgz
61 tar -xzf /tmp/rally.tgz -C ~/rally
62 echo "$GIT_SHA" > ~/rally/GIT_SHA
63
64 # 3) Build the env + native WASB (provider-agnostic DO scripts, reused on GCP per
deploy/gcp/README §4).
65 sudo bash deploy/digitalocean/mount_scratch.sh 2>/dev/null || true
66 export TMPDIR=/scratch 2>/dev/null || true
67 ./deploy/digitalocean/bootstrap.sh --gpu
68 . .venv/bin/activate
69 pip install -e '[storage-gcs]'
70 bash deploy/digitalocean/gpu_setup.sh || echo "WARN: gpu_setup (WASB env) returned
nonzero ? continuing"
71
72 # 4) Stage the WASB checkpoint (gpu_setup does NOT fetch weights ? README §4).
73 if [ -n "$WEIGHTS_URI" ]; then
74     mkdir -p ~/models/WASB-SBDT/pretrained_weights
75     gcloud storage cp "$WEIGHTS_URI" ~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar
76     export WASB_WEIGHTS=~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
77     fi
78
79 # 5) Run the regression on the gs:// golden clips in the committed manifest. Cost = VM
uptime so far.
80 UPTIME=$(( $(date +%s) - BOOT_EPOCH ))
81 EMAIL_FLAG=""; [ "$DO_EMAIL" = "true" ] && EMAIL_FLAG="--email --email-to $EMAIL_TO"
82 set +e
83 NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
84     python scripts/nightly_reelcount.py --vm-uptime-seconds "$UPTIME" $EMAIL_FLAG
85 RUN_RC=$?
86 set -e

```

```

87     echo "runner exit code: $RUN_RC  (0 pass / 1 regression / 4 stale)"
88
89     # 6) Upload the dated report to GCS for the record (alongside the email).
90     if [ -n "$BUCKET" ] && [ -d output/nightly_reelcount ]; then
91         gcloud storage cp -r "output/nightly_reelcount/$DATE."*
92         "gs://$BUCKET/$REPORT_PREFIX/$DATE/" || \
93         gcloud storage cp -r output/nightly_reelcount "gs://$BUCKET/$REPORT_PREFIX/$DATE/"
94     || true
95     fi
96
97     echo "==== nightly-regression done rc=$RUN_RC $(date -u +%FT%TZ) ====="
98     trap - ERR # disable the failure-email trap; the EXIT (self-delete) trap stays
99     # the EXIT trap deletes the VM

```

6. user (2026-06-22T09:42:17.827Z)

```

1     #!/usr/bin/env bash
2     # Launch ONE nightly engine-regression run on an ephemeral GCP GPU VM, then return. The
3     VM runs
4     # run_on_vm.sh as its startup-script (regression ? email ? SELF-DELETE), so this just
5     stages the repo
6     # and kicks the VM off. Driver-agnostic: call it from Cloud Scheduler?Cloud Run job, a
7     cron on any
8     # always-on box, the owner's machine, or a Claude routine. Needs gcloud auth + the repo
9     checkout.
10    #
11    # bash deploy/gcp/nightly_regression/launch.sh # L4, stages HEAD,
12    kicks the VM
13    # RALLY_GPU=t4 RALLY_REF=$(git rev-parse HEAD) bash
14    deploy/gcp/nightly_regression/launch.sh
15    #
16    # ? Run this MANUALLY once as a dry-run (watch the serial log) before wiring a scheduler
17    ? the VM-side
18    # flow is NOT covered by CI. Tail: gcloud compute instances get-serial-port-output
19    <name> --zone=<z>.
20    set -uo pipefail
21
22    PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
23    BUCKET="${RALLY_BUCKET:-khelsutra-rally-corpus}"
24    SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"
25    GPU="${RALLY_GPU:-l4}"
26    NAME="${RALLY_VM_NAME:-rally-nightly-$(date -u +%Y%m%d)}"
27    REF="${RALLY_REF:-HEAD}"
28    WEIGHTS_URI="${RALLY_WEIGHTS_URI:-gs://$BUCKET/weights/wasb_badminton_best.pth.tar}"
29    REPORT_PREFIX="${RALLY_REPORT_PREFIX:-nightly/reports}"
30    DO_EMAIL="${RALLY_EMAIL:-true}"
31    SMTP_SECRET="${NIGHTLY_SMTP_SECRET:-projects/$PROJECT/secrets/nightly-smtp-
32    pass/versions/latest}"
33    SMTP_USER="${NIGHTLY_SMTP_USER:-}"
34    EMAIL_TO="${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
35    HERE="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
36    STARTUP="$HERE/run_on_vm.sh"
37    export CLOUDSDK_CORE_DISABLE_PROMPTS=1
38
39    ZONES="${RALLY_GCP_ZONES:-asia-south1-a asia-south1-b asia-south1-c asia-southeast1-a
40    asia-southeast1-b asia-southeast1-c}"
41    if [ "$GPU" = "l4" ]; then MACHINE="g2-standard-4"; ACCEL=(); else
42    MACHINE="n1-standard-8"; ACCEL=(--accelerator=type=nvidia-tesla-t4,count=1); fi
43
44    # 1) Stage the repo @ REF to GCS (no GitHub auth on the VM; the worker SA can read the
45    bucket).
46    SHA=$(git rev-parse "$REF")
47    TGZ_URI="gs://$BUCKET/nightly/repo/${SHA}.tgz"
48    echo "== staging repo $SHA -> $TGZ_URI =="

```

```

37     TMP_TGZ="$(mktemp --suffix=.tgz)"
38     git archive --format=tar.gz -o "$TMP_TGZ" "$REF"
39     gcloud storage cp "$TMP_TGZ" "$TGZ_URI" --project="$PROJECT"
40     rm -f "$TMP_TGZ"
41
42     # 2) Zone-sweep create the VM (mirrors create_gpu_vm.sh) with run_on_vm.sh + the config
metadata.
43     ERR="$(mktemp)"
44     for Z in $ZONES; do
45         echo "== creating $NAME ($GPU) in $Z =="
46         if gcloud compute instances create "$NAME" --project="$PROJECT" --zone="$Z" \
47             --machine-type="$MACHINE" "${ACCEL[@]}" \
48             --maintenance-policy=TERMINATE --provisioning-model=STANDARD \
49             --image-family=common-cu129-ubuntu-2204-nvidia-580 --image-project=deeplearning-
platform-release \
50             --boot-disk-size=120GB --boot-disk-type=pd-balanced \
51             --service-account="$SA" --scopes=cloud-platform \
52             --metadata-from-file=startup-script="$STARTUP" \
53             --metadata=repo-tgz-uri="$TGZ_URI",git-sha="$SHA",gcs-bucket="$BUCKET",report-
prefix="$REPORT_PREFIX",weights-uri="$WEIGHTS_URI",email="$DO_EMAIL",smtp-
secret="$SMTP_SECRET",smtp-user="$SMTP_USER",email-to="$EMAIL_TO" \
54             2>"$ERR"; then
55         echo "? launched $NAME in $Z ? it will run the regression, email $EMAIL_TO, and
self-delete."
56         echo "    serial log: gcloud compute instances get-serial-port-output $NAME --zone=$Z
--project=$PROJECT"
57         echo "    if it doesn't self-delete: gcloud compute instances delete $NAME --zone=$Z
-q"
58         rm -f "$ERR"; exit 0
59     fi
60     if grep -qiE "STOCKOUT|does not have enough" "$ERR"; then echo "    (no $GPU capacity in
$Z ? next zone)"; continue; fi
61     echo "    create failed:"; tail -4 "$ERR"; rm -f "$ERR"; exit 1
62 done
63 echo "? no $GPU capacity in any swept zone ($ZONES). Retry later or RALLY_GPU=t4."
64 rm -f "$ERR"; exit 1
65

```

7. user (2026-06-22T09:42:18.565Z)

```

1     #!/usr/bin/env bash
2     # Wire the nightly engine-regression as a CLOUD-NATIVE, scale-to-zero job:
3     #   Cloud Scheduler (cron) ? Cloud Run JOB (the launcher image) ? launch.sh
4     #   ? ephemeral GPU VM (regression + email + SELF-DELETE).
5     # The only always-on cost is $0 (Cloud Run jobs + Scheduler are scale-to-zero; the GPU
VM is ephemeral).
6     #
7     # ? OWNER-RUN, ONCE, after a successful manual `bash
deploy/gcp/nightly_regression/launch.sh` dry-run.
8     # NOT verified in CI (no GCP/GPU here). Prereqs: GPU quota (README §2), the golden clips
+ weights in
9     # gs://<bucket>, and the SMTP secret (see README "Enable").
10    #
11    #   bash deploy/gcp/nightly_regression/register_scheduler.sh           # build + deploy
+ schedule (03:00 IST)
12    #   RALLY_CRON='0 3 * * *' RALLY_TZ='Asia/Kolkata' bash ../register_scheduler.sh
13    set -euo pipefail
14
15    PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
16    REGION="${RALLY_RUN_REGION:-asia-south1}"
17    JOB="${RALLY_RUN_JOB:-nightly-regression-launcher}"
18    SCHED="${RALLY_SCHED_NAME:-nightly-regression}"
19    CRON="${RALLY_CRON:-0 3 * * *}"           # 03:00 daily
20    TZ="${RALLY_TZ:-Asia/Kolkata}"
21    SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"

```

```

22     IMAGE="${RALLY_LAUNCHER_IMAGE:-${REGION}-docker.pkg.dev/${PROJECT}/rally/nightly-
launcher:latest}"
23     REPO_ROOT="$(git rev-parse --show-toplevel)"
24     export CLOUDSDK_CORE_DISABLE_PROMPTS=1
25
26     echo "== enable APIs (run, scheduler, build, artifact registry) =="
27     gcloud services enable run.googleapis.com cloudscheduler.googleapis.com \
28         cloudbuild.googleapis.com artifactregistry.googleapis.com --project="$PROJECT"
29
30     echo "== ensure the Artifact Registry repo 'rally' exists =="
31     gcloud artifacts repositories describe rally --location="$REGION" --project="$PROJECT"
>/dev/null 2>&1 || \
32     gcloud artifacts repositories create rally --repository-format=docker
--location="$REGION" --project="$PROJECT"
33
34     echo "== build + push the launcher image from the repo root =="
35     ( cd "$REPO_ROOT" && gcloud builds submit --project="$PROJECT" --tag "$IMAGE" \
36         --gcs-source-staging-dir="gs://${RALLY_BUILD_BUCKET:-khealsutra-rally-
corpus}/cloudbuild" \
37         -f deploy/gcp/nightly_regression/Dockerfile.launcher . )
38
39     echo "== create/update the Cloud Run JOB (launcher needs compute-admin to create the GPU
VM) =="
40     gcloud run jobs deploy "$JOB" --project="$PROJECT" --region="$REGION" \
41         --image="$IMAGE" --service-account="$SA" --max-retries=0 --task-timeout=900s \
42         --set-env-vars="RALLY_GPU=${RALLY_GPU:-
l4},RALLY_EMAIL=true,NIGHTLY_EMAIL_TO=${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
43
44     echo "== schedule it (Cloud Scheduler ? Run job admin API) =="
45     RUN_URI="https://${REGION}-
run.googleapis.com/apis/run.googleapis.com/v1/namespaces/${PROJECT}/jobs/${JOB}:run"
46     gcloud scheduler jobs describe "$SCHED" --location="$REGION" --project="$PROJECT"
>/dev/null 2>&1 && \
47     gcloud scheduler jobs delete "$SCHED" --location="$REGION" --project="$PROJECT"
--quiet || true
48     gcloud scheduler jobs create http "$SCHED" --project="$PROJECT" --location="$REGION" \
49         --schedule="$CRON" --time-zone="$TZ" \
50         --uri="$RUN_URI" --http-method=POST \
51         --oauth-service-account-email="$SA"
52
53     cat <<EOF
54
55     ? Scheduled '$SCHED' ($CRON $TZ) ? Cloud Run job '$JOB' ? launch.sh ? ephemeral GPU VM.
56     IAM the launcher SA needs (grant once if missing):
57         roles/compute.instanceAdmin.v1 (create/delete the GPU VM)
58         roles/iam.serviceAccountUser (attach $SA to the VM)
59         roles/run.invoker + roles/cloudscheduler.jobRunner (Scheduler ? Run)
60     Trigger now to test: gcloud run jobs execute $JOB --region=$REGION
--project=$PROJECT
61     Remove: gcloud scheduler jobs delete $SCHED --location=$REGION; gcloud
run jobs delete $JOB --region=$REGION
62     EOF
63

```

8. assistant (2026-06-22T09:42:20.816Z)

Now let me read the remaining files:

9. user (2026-06-22T09:42:22.195Z)

```

1     # Tiny launcher image for the nightly engine-regression Cloud Run JOB. It is NOT the GPU
workload ?
2     # it only stages the repo + creates the ephemeral GPU VM (which self-deletes). Scale-to-
zero: the job
3     # runs for a few seconds nightly, then exits. Base = google/cloud-sdk (gcloud + git

```

```

preinstalled).
4     FROM gcr.io/google.com/cloudsdktool/cloud-sdk:slim
5     WORKDIR /app
6     # The repo is the build context (run `gcloud builds submit` from the repo root, see
register_scheduler.sh).
7     COPY . /app
8     ENTRYPOINT ["bash", "deploy/gcp/nightly_regression/launch.sh"]
9

```

10. user (2026-06-22T09:42:23.138Z)

```

1     {
2         "_doc": "Nightly E2E engine reel-count manifest (scripts/nightly_reelcount.py). The
configured+checkpointed pipeline runs on each video on a GCP GPU VM; the reel count + quality
metrics are asserted vs eval_baselines/reelcount_baseline.json. ADD A VIDEO = append one row +
re-freeze: `python scripts/nightly_reelcount.py --update-baseline` (commit the updated
baseline). URIs may be gs:// (recommended, co-located with the VM), gdrive://, or local paths.
labels_uri (optional) is a golden-features JSON (gts key) or a start,end CSV ? present =>
quality metrics (tol_f1/f1@0.5/f1@0.25) are scored too. EDIT the example rows below to your
actual short (~5-6 min) OWNED, minors-free clips before enabling.",
3         "segmenter": "wasb_hybrid",
4         "detector_impl": "native",
5         "config_overrides": {
6             "indexing.default_segmenter": "wasb_hybrid",
7             "indexing.detector_impl": "native"
8         },
9         "machine_usd_per_hr": 0.70,
10        "fx_inr_per_usd": 83.0,
11        "videos": [
12            {
13                "name": "GX030094",
14                "uri": "gs://khelsutra-rally-corpus/nightly/videos/GX030094.mp4",
15                "labels_uri": "gs://khelsutra-rally-
corpus/nightly/labels/GX030094_golden_features.json"
16            },
17            {
18                "name": "Badminton_BXH_2",
19                "uri": "gs://khelsutra-rally-corpus/nightly/videos/Badminton_BXH_2.mp4",
20                "labels_uri": "gs://khelsutra-rally-
corpus/nightly/labels/Badminton_BXH_2_golden_features.json"
21            },
22            {
23                "name": "GX010128",
24                "uri": "gs://khelsutra-rally-corpus/nightly/videos/GX010128.mp4",
25                "labels_uri": "gs://khelsutra-rally-
corpus/nightly/labels/GX010128_golden_features.json"
26            }
27        ]
28    }
29

```

11. assistant (2026-06-22T09:42:25.775Z)

Now let me examine the billing module and run some targeted searches for critical issues:

12. user (2026-06-22T09:42:27.276Z)

```

C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\nightly-e2e/backend/billing.py

```

13. user (2026-06-22T09:42:29.113Z)

```

Found 5 files
scripts\nightly_reelcount.py

```

```
tests\test_billing.py
backend\main.py
backend\infrastructure\database.py
backend\billing.py
```

14. assistant (2026-06-22T09:42:31.394Z)

Let me read the billing module to understand the cost calculation:

15. user (2026-06-22T09:42:31.829Z)

```
1      """Per-job cost ledger ? pure cost math (COMPUTE_DECOUPLED_SERVING M8, decision D8).
2
3      ``cost_usd(job) = ?_resource usage(job, resource) × rate(resource, pricebook_version)``
4      (the formula from the archived 05-COST-ATTRIBUTION design). Cost is computed + stored in
5      **USD**
6      (matches GCP billing = one source of truth) and **displayed in INR** at a configurable
7      FX rate
8      (the Bangalore market). The pricebook is a versioned DB table
9      (``backend/infrastructure/database.py``);
10     this module is the pure arithmetic over a resolved ``{resource: unit_price_usd}`` rates
11     dict, so it
12     is unit-tested with zero IO and the DB layer owns the version/gpu_type lookup.
13
14     **Default-OFF:** with the seeded ``placeholder-v0`` pricebook (every rate ``0.0``) every
15     cost is
16     ``0.0`` ? nothing user-facing changes until the owner inserts a real, calibrated
17     pricebook version.
18     """
19
20     from __future__ import annotations
21
22     from typing import Dict, Mapping, Tuple
23
24     # Metered resource keys ? shared by the usage dict, the pricebook rows, and the
25     breakdown.
26     GPU_SECONDS = "gpu_seconds"      # GPU-active seconds (the big one) ? rate is per-second,
27     per gpu_type
28     WALL_SECONDS = "wall_seconds"    # total wall time (separate from GPU so idle/sync isn't
29     billed as GPU)
30     EGRESS_GB = "egress_gb"          # reel/byte egress in GB
31     GEMINI_USD = "gemini_usd"        # provider-reported cost, ALREADY in USD (a passthrough,
32     not rate-×)
33
34     # Resources whose cost = amount × unit_price (GEMINI_USD is excluded ? it is already a
35     USD amount).
36     _RATE_MULTIPLIED = (GPU_SECONDS, WALL_SECONDS, EGRESS_GB)
37
38     def compute_cost_usd(usage: Mapping[str, float],
39                          rates: Mapping[str, float]) -> Tuple[float, Dict[str, float]]:
40         """Return ``(cost_usd, breakdown)`` for one job.
41
42         ``usage`` maps a resource key to its measured amount (seconds / GB); ``rates`` maps
43         a resource
44         key to its USD unit price (already resolved for this pricebook version + gpu_type by
45         the DB
46         layer). ``GEMINI_USD`` in ``usage`` is added verbatim (it is already a USD cost, not
47         multiplied).
48         Missing amounts/rates default to ``0.0`` ? with the placeholder pricebook the total
49         is ``0.0``.
50         """
51         breakdown: Dict[str, float] = {}
52         total = 0.0
53         for res in _RATE_MULTIPLIED:
```

```

40         amount = float(usage.get(res, 0.0) or 0.0)
41         rate = float(rates.get(res, 0.0) or 0.0)
42         cost = round(amount * rate, 6)      # round the LINE, then sum rounded lines so
the
43         if amount or rate:                  # breakdown reconciles to the total exactly
(no drift),
44         breakdown[res] = cost               # and record the line even at $0 for
transparency
45         total += cost
46         gemini = round(float(usage.get(GEMINI_USD, 0.0) or 0.0), 6)
47         if gemini:
48             breakdown[GEMINI_USD] = gemini
49         total += gemini
50         return round(total, 6), breakdown
51
52
53     def apply_margin(cost_usd: float, margin: float) -> float:
54         """Resale price = `cost_usd * (1 + margin)` . `margin` is clamped ? 0 (never
below cost)."""
55         return round(cost_usd * (1.0 + max(0.0, float(margin))), 6)
56
57
58     def to_inr(cost_usd: float, fx_inr_per_usd: float) -> float:
59         """Display conversion: USD ? INR at the configured FX rate (2 dp, the display
currency)."""
60         return round(float(cost_usd) * float(fx_inr_per_usd), 2)
61

```

16. assistant (2026-06-22T09:42:37.958Z)

Now let me search for critical patterns and edge cases in the main script:

17. user (2026-06-22T09:42:39.584Z)

```

24-Freeze the baseline from the current configured+checkpointed engine (after an intended
change, or first run):
25- python scripts/nightly_reelcount.py --update-baseline
26-
27:Nightly check (exit 1 on regression; writes output/nightly_reelcount/<date>.md; emails if
--email):
28- python scripts/nightly_reelcount.py --email
29-
30-Exit codes: 0 pass · 1 regression (reel-count changed / a video errored / quality dropped) ·
2 operational
--
156-         return "STALE"
157-         return "PASS"
158-
159:     def exit_code(self) -> int:
160-         if self.regressions:
161-             return 1
162-         if self.stale:
163-             return 1
164-         return 0
165-
166-
167-
168-
169-
170-
171-
172-
173-
174-
175-
176-
177-
178-
179-
180-
181-
182-
183-
184-
185-
186-
187-
188-
189-
190-
191-
192-
193-
194-
195-
196-
197-
198-
199-
200-
201-
202-
203-
204-
205-
206-
207-
208-
209-
210-
211-
212-
213-
214-
215-
216-
217-
218-
219-
220-
221-
222-
223-
224-
225-
226-
227- L: List[str] = [
228-     f"# Nightly engine regression (reel-count + quality + cost) ? {date}",
229-     "",
230-     f"*Status: {badge}** · exit code {result.exit_code()} · segmenter
`{current.get('segmenter')}`",
231-     "",
232-     f"- HEAD `{head}` · baseline `{baseline.get('git_commit', '?')}` (frozen
{baseline.get('generated_at', '?')}")",
233-     f"- **Run cost: ${cost.get('usd', 0):.4f} ({cost.get('inr', 0):.2f})** · "
234-     ""
235- ]
236-
237-
238-
239-
240-
241-
242-
243-
244-
245-
246-
247-
248-
249-
250-
251-
252-
253-
254-
255-
256-
257-
258-
259-
260-
261-
262-
263-
264-
265-
266-
267-
268-
269-
270-
271-
272-
273-
274-
275-
276-
277-
278-
279-
280-
281-
282-
283-
284-
285-
286-
287- for row in csv.reader(fh):
288-     try:

```

```

289-         out.append((float(row[0]), float(row[1])))
290-     except (ValueError, IndexError):
291-         continue
292- return out
293-
--
327- name = video["name"]
328- try:
329-     local = fetch(video["uri"])
330- except Exception as e: # noqa: BLE001 ? surface, never hide
331-     return {"name": name, "reel_count": None, "ok": False, "error": f"fetch failed:
{e}",
332-            "quality": None, "wall_seconds": 0.0}
333-
--
411-         return
412-     try:
413-         setattr(obj, parts[-1], value)
414-     except Exception: # noqa: BLE001 ? overrides are best-effort; a bad key shouldn't crash
the run
415-         pass
416-
417-
--
421-         out = subprocess.run(["git", "-C", REPO_ROOT, "rev-parse", "--short", "HEAD"],
422-                               capture_output=True, text=True, timeout=10)
423-         return out.stdout.strip() or "?"
424-     except Exception: # noqa: BLE001
425-         return "?"
426-
427-
--
484-
485-     try:
486-         run_results, total_wall = run_corpus(manifest, baseline, rerun_on_mismatch=not
args.no_rerun_guard)
487-     except Exception as e: # noqa: BLE001 ? operational failure (deps/config) ? exit 2,
never a false "pass"
488-         log.error("run failed: %s", e)
489-         return 2
490-     if not run_results:
--
523-         with open(md_path, "w", encoding="utf-8") as fh:
524-             fh.write(report + "\n")
525-         with open(os.path.join(args.report_dir, f"{date}.json"), "w", encoding="utf-8") as
fh:
526-             json.dump({"date": date, "status": result.overall_status(), "exit_code":
result.exit_code(),
527-                       "snapshot": snapshot, "regressions": [f.message() for f in
result.regressions]},
528-                       fh, indent=2, sort_keys=True)
529-         log.info("report ? %s", md_path)
--
536-         subject = f"[KhelSutra nightly] {result.overall_status()} ? reel-count engine
regression {date}"
537-         sent = send_report(subject, report, to=args.email_to)
538-         log.info("email %s", "sent" if sent else "skipped (no creds ? report still
written)")
539-     except Exception as e: # noqa: BLE001 ? a broken mailer must NOT mask the
regression verdict
540-         log.error("email failed (report still written): %s", e)
541-
542-     code = result.exit_code()
543-     if code == 1:
544-         log.error("REEL-COUNT REGRESSION (%d) ? see %s", len(result.regressions),

```

```

args.report_dir)
545-     elif code == 4:
546-         log.warning("baseline STALE ? corpus/segmenter changed; re-freeze with --update-
baseline")
547-     else:
548-         log.info("nightly reel-count check PASSED (cost $%.4f / ?%.2f)", cost["usd"],
cost["inr"])
549:     return code
550-
551-
552-if __name__ == "__main__":
553:    raise SystemExit(main())

```

18. assistant (2026-06-22T09:42:41.631Z)

Let me check the re-run guard logic more carefully and look for edge cases:

19. user (2026-06-22T09:42:43.905Z)

```

305-         out.append((float(start), float(end)))
306-     return out
307-
308-
309-def run_one_video(video: Dict[str, Any], segmenter: str, config: Any,
310:                 rerun_on_mismatch: bool = True,
311:                 baseline_count: Optional[int] = None) -> Dict[str, Any]:
312-     """Run the configured pipeline on one video ? reel count (+ quality vs labels). Lazy
backend imports.
313-
314-     The re-run-once guard (the one known non-determinism ? a transient
``add_play_segment``?None drop,
315-     database.py): if the count != the frozen baseline, re-process the video ONCE; a
transient drop won't
316-     reproduce, a real change will."""
317-     --
357-         total_wall += wall
358-         if err is not None:
359-             return {"name": name, "reel_count": None, "ok": False, "error": err,
360-                    "quality": None, "wall_seconds": round(total_wall, 2)}
361-         # re-run-once guard for the transient DB-drop flake
362-         if rerun_on_mismatch and baseline_count is not None and count != baseline_count:
363-             count2, intervals2, wall2, err2 = _process(os.path.join(td, "run2.db"))
364-             total_wall += wall2
365-             if err2 is None and count2 is not None:
366-                 count, intervals = count2, intervals2 # the reproduced (or corrected) value
367-
368-         quality = None
369-         if gts and intervals is not None:
370-             row = rally_seg_eval.score_intervals(intervals, gts)
371-             quality = {k: row[k] for k in ("tol_f1", "f1@0.5", "f1@0.25", "tol_precision",
"tol_recall")}
372-         --
379-         with open(path, encoding="utf-8") as fh:
380-             return json.load(fh)
381-
382-
383-def run_corpus(manifest: Dict[str, Any], baseline: Optional[Dict[str, Any]],
384:              rerun_on_mismatch: bool = True) -> Tuple[List[Dict[str, Any]], float]:
385-     """Run every manifest video through the configured pipeline. Returns (run_results,
total_wall_s)."""
386-     from backend.config import load_config
387-
388-     config = load_config()
389-     # apply manifest config overrides onto the typed config (dotted keys)
390-     --

```

```

393-     b_pv = (baseline or {}).get("per_video", {})
394-     results: List[Dict[str, Any]] = []
395-     total_wall = 0.0
396-     for v in manifest.get("videos", []):
397-         base_count = (b_pv.get(v["name"]) or {}).get("reel_count")
398-         r = run_one_video(v, segmenter, config, rerun_on_mismatch, base_count)
399-         total_wall += float(r.get("wall_seconds") or 0.0)
400-         results.append(r)
401-     return results, total_wall
402-
403-
--
481-         return 2
482-     manifest = load_manifest(args.manifest)
483-     baseline = load_baseline(args.baseline)
484-
485-     try:
486-         run_results, total_wall = run_corpus(manifest, baseline, rerun_on_mismatch=not
args.no_rerun_guard)
487-     except Exception as e: # noqa: BLE001 ? operational failure (deps/config) ? exit 2,
never a false "pass"
488-         log.error("run failed: %s", e)
489-         return 2
490-     if not run_results:
491-         log.error("no videos in manifest %s", args.manifest)

```

20. assistant (2026-06-22T09:42:46.032Z)

Now let me check for the comparison logic and quality tolerance behavior:

21. user (2026-06-22T09:42:48.609Z)

```

13-     **EXACT** (with a re-run-once guard for the one known flake: a transient DB-insert-None,
see $guard).
14-     ? QUALITY METRICS per video (when golden labels are provided) ? the R2 tolerance-match
block from
15-     ``backend.eval.rally_seg_eval`` (tol_f1 / f1@0.5 / f1@0.25), compared with a small
tolerance.
16-     ? COST of the run ? VM uptime (or processing wall) × machine $/hr via ``backend.billing`` ?
USD + INR.
17-
--
53-
54-#: Quality metrics we report + (when labels exist) gate ? higher is better, small tolerance.
55:QUALITY_HIGHER: Tuple[str, ...] = ("tol_f1", "f1@0.5", "f1@0.25")
56:DEFAULT_QUALITY_TOL = 0.02 # quality is float/labeled ? a looser tripwire than the exact
reel-count
57:DEFAULT_FX_INR_PER_USD = 83.0
--
62:@dataclass(frozen=True)
63:class Tolerances:
64-     quality_tol: float = DEFAULT_QUALITY_TOL
65-
66-     def to_dict(self) -> Dict[str, Any]:
67-         return {"quality_tol": self.quality_tol}
68-
69-
--
98-     """Compact, comparable snapshot from per-video run results.
99-
100-     Each run result: ``{name, reel_count, ok, error, quality: {tol_f1,...}|None,
wall_seconds}``.
101-     A video that errored keeps ``reel_count=None`` + ``ok=False`` (so the diff flags it,
never hides it)."""
102-     per_video: Dict[str, Any] = {}

```

```

--
171-     * Corpus (video set) changed ? STALE + report added/removed; still diff the
INTERSECTION.
172-     * Per shared video: reel_count EXACT (any change = regression); a pipeline ERROR =
regression;
173:     quality metrics drop beyond ``quality_tol`` = regression (rise = improvement).
174-     """
175-     res = NightlyResult()
--
203-         if bv is None or cv is None:
204-             continue
205:         if cv < bv - tols.quality_tol:
206-             res.findings.append(Finding(v, m, "regression", bv, cv))
207:         elif cv > bv + tols.quality_tol:
208-             res.findings.append(Finding(v, m, "improvement", bv, cv))
209-     return res
--
244-     b_pv, c_pv = baseline.get("per_video", {}), current.get("per_video", {})
245-     L += ["## Per-video", "",
246:         "| video | reels (base?now) | tol_f1 (base?now) | f1@0.5 | status |",
247:         "|---|---|---|---|---|"]
248-     for v in sorted(set(b_pv) | set(c_pv)):
--
252-         rc = f"'?' if bc is None else bc}{'ERROR' if not c.get('ok', True) else ('?' if cc
is None else cc)}"
253-         bq, cq = b.get("quality"), c.get("quality")
254:         tf = f"_{q(bq, 'tol_f1')}?_{q(cq, 'tol_f1')}"
255:         f5 = f"_{q(bq, 'f1@0.5')}?_{q(cq, 'f1@0.5')}"
256-         vstatus = "?" if any(f.video == v and f.kind == "regression" for f in
result.findings) else ""
257-         L.append(f"| {v} | {rc} | {tf} | {f5} | {vstatus} |")
--
369-         if gts and intervals is not None:
370-             row = rally_seg_eval.score_intervals(intervals, gts)
371:             quality = {k: row[k] for k in ("tol_f1", "f1@0.5", "f1@0.25", "tol_precision",
"tol_recall")}
372-                 if k in row}
373-
--
474-     log = logging.getLogger("nightly_reelcount")
475-     args = build_parser().parse_args(argv)
476:     tols = Tolerances(quality_tol=args.quality_tol)
477-     date = args.date or _dt.date.today().isoformat()
478-

```

22. assistant (2026-06-22T09:42:50.687Z)

Let me check the test files to better understand the expected behavior:

23. user (2026-06-22T09:42:51.826Z)

tests\test_nightly_regression.py
tests\test_nightly_reelcount.py

24. user (2026-06-22T09:42:54.173Z)

```

1     """Tests for scripts/nightly_reelcount.py + scripts/nightly_email.py ? the pure cores.
2
3     Fully offline: no GPU, no video, no network, no real corpus. The comparison / cost /
report / email
4     formatting operate on plain dicts + strings, so reel-count regressions, quality drops,
pipeline
5     errors, corpus drift, and the no-creds email skip are all exercised deterministically.
The actual

```

```

6     pipeline run (run_one_video / run_corpus) needs a video + GPU and is owner-verified on
the VM.
7     """
8     import importlib.util
9     import os
10    import sys
11
12    import pytest
13
14    _SCRIPTS = os.path.join(os.path.dirname(os.path.dirname(os.path.abspath(__file__))),
"scripts")
15
16
17    def _load(modname):
18        spec = importlib.util.spec_from_file_location(modname, os.path.join(_SCRIPTS,
f"{modname}.py"))
19        mod = importlib.util.module_from_spec(spec)
20        sys.modules[modname] = mod # register before exec so dataclass annotations resolve
21        spec.loader.exec_module(mod)
22        return mod
23
24
25    nr = _load("nightly_reelcount")
26    ne = _load("nightly_email")
27
28
29    # ----- #
30    # Builders
31    # ----- #
32    def _video(name, reel_count, ok=True, error=None, quality=None):
33        return {"name": name, "reel_count": reel_count, "ok": ok, "error": error,
34                "quality": quality, "wall_seconds": 1.0}
35
36
37    def _snap(per_video, segmenter="wasb_hybrid"):
38        return {"segmenter": segmenter, "per_video": per_video,
39                "cost": {"usd": 0.05, "inr": 4.15, "billed_seconds": 500,
"machine_usd_per_hr": 0.35}}
40
41
42    TOLS = nr.Tolerances()
43
44
45    # ----- #
46    # cost_report (billing wrapper)
47    # ----- #
48    def test_cost_report_math():
49        c = nr.cost_report(billed_seconds=3600, machine_usd_per_hr=0.35,
fx_inr_per_usd=83.0)
50        assert c["usd"] == pytest.approx(0.35, abs=1e-4) # 1hr @ $0.35
51        assert c["inr"] == pytest.approx(0.35 * 83.0, abs=0.01)
52        assert c["billed_seconds"] == 3600
53
54
55    def test_cost_report_zero_for_zero_time():
56        c = nr.cost_report(billed_seconds=0, machine_usd_per_hr=0.35, fx_inr_per_usd=83.0)
57        assert c["usd"] == 0.0 and c["inr"] == 0.0
58
59
60    # ----- #
61    # summarize_results
62    # ----- #
63    def test_summarize_results():
64        rr = [_video("v1", 5), _video("v2", 3, quality={"tol_f1": 0.4})]
65        snap = nr.summarize_results(rr, "wasb_hybrid", {"usd": 0.1})

```

```

66         assert snap["n"] == 2
67         assert snap["per_video"]["v1"]["reel_count"] == 5
68         assert snap["per_video"]["v2"]["quality"]["tol_f1"] == 0.4
69
70
71     # ----- #
72     # compare ? reel count is EXACT
73     # ----- #
74     def test_identical_counts_pass():
75         base = _snap({"v1": {"reel_count": 5, "ok": True}})
76         cur = _snap({"v1": {"reel_count": 5, "ok": True}})
77         res = nr.compare(base, cur, TOLS)
78         assert res.overall_status() == "PASS" and res.exit_code() == 0
79
80
81     def test_reel_count_change_is_regression():
82         base = _snap({"v1": {"reel_count": 5, "ok": True}})
83         cur = _snap({"v1": {"reel_count": 4, "ok": True}})          # one reel vanished
84         res = nr.compare(base, cur, TOLS)
85         assert res.exit_code() == 1
86         r = res.regressions[0]
87         assert r.metric == "reel_count" and r.baseline == 5 and r.current == 4
88
89
90     def test_reel_count_increase_is_also_regression():
91         # ANY change to the count is a regression ? the count must be stable, either
92         direction.
93         base = _snap({"v1": {"reel_count": 5, "ok": True}})
94         cur = _snap({"v1": {"reel_count": 6, "ok": True}})
95         assert nr.compare(base, cur, TOLS).exit_code() == 1
96
97     def test_pipeline_error_is_regression():
98         base = _snap({"v1": {"reel_count": 5, "ok": True}})
99         cur = _snap({"v1": {"reel_count": None, "ok": False, "error": "WASB OOM"}})
100         res = nr.compare(base, cur, TOLS)
101         assert res.exit_code() == 1
102         assert res.regressions[0].metric == "error" and "WASB OOM" in
103         res.regressions[0].detail
104
105     # ----- #
106     # compare ? quality metrics use tolerance
107     # ----- #
108     def test_quality_drop_beyond_tol_is_regression():
109         base = _snap({"v1": {"reel_count": 5, "ok": True, "quality": {"tol_f1": 0.40}}})
110         cur = _snap({"v1": {"reel_count": 5, "ok": True, "quality": {"tol_f1": 0.30}}})  #
111         ?0.10 ? tol
112         res = nr.compare(base, cur, TOLS)
113         assert res.exit_code() == 1 and res.regressions[0].metric == "tol_f1"
114
115     def test_quality_drop_within_tol_passes():
116         base = _snap({"v1": {"reel_count": 5, "ok": True, "quality": {"tol_f1": 0.40}}})
117         cur = _snap({"v1": {"reel_count": 5, "ok": True, "quality": {"tol_f1": 0.39}}})  #
118         within 0.02
119         assert nr.compare(base, cur, TOLS).exit_code() == 0
120
121     def test_quality_rise_is_improvement_not_regression():
122         base = _snap({"v1": {"reel_count": 5, "ok": True, "quality": {"f1@0.5": 0.40}}})
123         cur = _snap({"v1": {"reel_count": 5, "ok": True, "quality": {"f1@0.5": 0.55}}})
124         res = nr.compare(base, cur, TOLS)
125         assert res.exit_code() == 0
126         assert [f.metric for f in res.improvements] == ["f1@0.5"]

```

```

127
128
129 # ----- #
130 # compare ? staleness (segmenter / corpus drift)
131 # ----- #
132 def test_segmenter_change_is_stale():
133     base = _snap({"v1": {"reel_count": 5, "ok": True}}, segmenter="wasb_hybrid")
134     cur = _snap({"v1": {"reel_count": 5, "ok": True}}, segmenter="motion")
135     res = nr.compare(base, cur, TOLS)
136     assert res.overall_status() == "STALE" and res.exit_code() == 4
137
138
139 def test_corpus_drift_is_stale_but_checks_intersection():
140     base = _snap({"v1": {"reel_count": 5, "ok": True}, "v2": {"reel_count": 3, "ok":
True}})
141     cur = _snap({"v1": {"reel_count": 9, "ok": True}, "v3": {"reel_count": 1, "ok":
True}})
142     res = nr.compare(base, cur, TOLS)
143     assert "v3" in res.added and "v2" in res.removed
144     # v1 regressed on the intersection ? regression wins over stale (exit 1)
145     assert res.exit_code() == 1
146     assert any(f.video == "v1" and f.metric == "reel_count" for f in res.regressions)
147
148
149 def test_corpus_drift_clean_intersection_is_stale_exit_4():
150     base = _snap({"v1": {"reel_count": 5, "ok": True}, "v2": {"reel_count": 3, "ok":
True}})
151     cur = _snap({"v1": {"reel_count": 5, "ok": True}}) # v2 dropped, v1 unchanged
152     res = nr.compare(base, cur, TOLS)
153     assert res.exit_code() == 4 and res.overall_status() == "STALE"
154
155
156 # ----- #
157 # report formatting (smoke)
158 # ----- #
159 def test_format_report_has_status_cost_and_regression():
160     base = _snap({"v1": {"reel_count": 5, "ok": True, "quality": {"tol_f1": 0.4,
"f1@0.5": 0.5}}})
161     base["git_commit"] = "aaa"
162     base["generated_at"] = "2026-06-21"
163     cur = _snap({"v1": {"reel_count": 3, "ok": True, "quality": {"tol_f1": 0.4,
"f1@0.5": 0.5}}})
164     res = nr.compare(base, cur, TOLS)
165     rpt = nr.format_report(base, cur, res, "2026-06-22", "bbb")
166     assert "Nightly engine regression" in rpt
167     assert "REGRESSION" in rpt
168     assert "Run cost:" in rpt and "?" in rpt
169     assert "reel_count" in rpt
170     assert "v1" in rpt
171
172
173 def test_format_report_pass_badge():
174     base = _snap({"v1": {"reel_count": 5, "ok": True}})
175     cur = _snap({"v1": {"reel_count": 5, "ok": True}})
176     res = nr.compare(base, cur, TOLS)
177     assert "PASS" in nr.format_report(base, cur, res, "2026-06-22", "bbb")
178
179
180 # ----- #
181 # email ? build + creds resolution + graceful skip
182 # ----- #
183 def test_email_build_message():
184     msg = ne.build_message("subj", "body **md**", "from@x.com", "to@y.com")
185     assert msg["Subject"] == "subj" and msg["To"] == "to@y.com" and msg["From"] ==
"from@x.com"

```

```

186         assert "body **md**" in msg.get_content()
187
188
189     def test_email_resolve_secret_from_env():
190         assert ne.resolve_secret({"NIGHTLY_SMTP_PASS": "hunter2"}) == "hunter2"
191
192
193     def test_email_resolve_secret_none_when_absent():
194         assert ne.resolve_secret({}) is None
195
196
197     def test_email_send_skips_without_creds():
198         # No password resolvable ? graceful skip (False), never raises, no network.
199         assert ne.send_report("s", "b", to="x@y.com", env={"NIGHTLY_EMAIL_TO": "x@y.com"})
is False
200
201
202     def test_tolerances_default():
203         assert nr.Tolerances().quality_tol == 0.02
204
205
206     def test_segments_to_intervals_both_key_conventions():
207         # motion/DB shape (start_time/end_time) AND the alternate (start/end) both parse;
bad rows skip.
208         segs = [{"start_time": 1.0, "end_time": 2.0}, {"start": 5.0, "end": 7.0}, {"nope":
1}]
209         assert nr._segments_to_intervals(segs) == [(1.0, 2.0), (5.0, 7.0)]
210         assert nr._segments_to_intervals([]) == []
211

```

25. assistant (2026-06-22T09:42:56.332Z)

Now let me examine the email implementation more closely for security issues:

26. user (2026-06-22T09:42:57.967Z)

```

4:dependency-light (stdlib ``smtplib`` + ``email.message``) and **graceful**: if no SMTP
password can
10: ? SMTP host: ``NIGHTLY_SMTP_HOST`` (default ``smtp.gmail.com``) . port
``NIGHTLY_SMTP_PORT`` (587, STARTTLS)
84: import smtplib
86: with smtplib.SMTP(host, port, timeout=60) as s:
88:     s.login(user, password)

```

27. assistant (2026-06-22T09:43:00.030Z)

Now let me check the startup script for more issues:

28. user (2026-06-22T09:43:01.440Z)

```

8:#         upload report to GCS ? SELF-DELETE (even on failure, via the EXIT trap).
31:# ALWAYS delete the VM on exit (success OR failure) ? never leave a billing GPU VM behind.
37:trap cleanup EXIT
52:trap 'fail_email' ERR
67:./deploy/digitalocean/bootstrap.sh --gpu
70:bash deploy/digitalocean/gpu_setup.sh || echo "WARN: gpu_setup (WASB env) returned nonzero ?
continuing"
82:set +e
85:RUN_RC=$?
86:set -e
87:echo "runner exit code: $RUN_RC (0 pass / 1 regression / 4 stale)"
95:echo "==== nightly-regression done rc=$RUN_RC $(date -u +%FT%TZ) ====="
96:trap - ERR # disable the failure-email trap; the EXIT (self-delete) trap stays
97:# the EXIT trap deletes the VM

```

29. assistant (2026-06-22T09:43:03.352Z)

Let me check the baseline.json existence and structure:

30. user (2026-06-22T09:43:04.317Z)

(Bash completed with no output)

31. assistant (2026-06-22T09:43:05.877Z)

Let me check if there's a baseline at all:

32. user (2026-06-22T09:43:06.866Z)

Exit code 2

```
/usr/bin/bash: eval: line 1: unexpected EOF while looking for matching `''
```

33. user (2026-06-22T09:43:09.908Z)

```
total 56
drwxr-xr-x 1 avidu 197609      0 Jun 22 15:05 .
drwxr-xr-x 1 avidu 197609      0 Jun 22 15:10 ..
-rw-r--r-- 1 avidu 197609 4696 Jun 22 14:36 experiment_leaderboard.json
drwxr-xr-x 1 avidu 197609      0 Jun 22 14:36 fixtures
-rw-r--r-- 1 avidu 197609 1977 Jun 22 14:36 gemini_wrapper_2026-06-10.json
-rw-r--r-- 1 avidu 197609  689 Jun 22 14:36 heuristic_lovo_n6.json
-rw-r--r-- 1 avidu 197609 17458 Jun 22 14:36 nightly_baseline.json
-rw-r--r-- 1 avidu 197609 1601 Jun 22 15:05 reelcount_manifest.json
```

34. assistant (2026-06-22T09:43:13.210Z)

Now let me check for a critical issue with the re-run guard and make sure I understand the control flow:

35. user (2026-06-22T09:43:15.095Z)

```
359-         return {"name": name, "reel_count": None, "ok": False, "error": err,
360-                 "quality": None, "wall_seconds": round(total_wall, 2)}
361-     # re-run-once guard for the transient DB-drop flake
362-     if rerun_on_mismatch and baseline_count is not None and count != baseline_count:
363-         count2, intervals2, wall2, err2 = _process(os.path.join(td, "run2.db"))
364-         total_wall += wall2
365-         if err2 is None and count2 is not None:
366-             count, intervals = count2, intervals2 # the reproduced (or corrected) value
367-
368-     quality = None
369-     if gts and intervals is not None:
370-         row = rally_seg_eval.score_intervals(intervals, gts)
371-         quality = {k: row[k] for k in ("tol_f1", "f1@0.5", "f1@0.25", "tol_precision",
"tol_recall")}
```

36. assistant (2026-06-22T09:43:25.286Z)

Perfect. Now let me review the compare logic once more to check for masking issues:

37. user (2026-06-22T09:43:26.580Z)

```
167     def compare(baseline: Dict[str, Any], current: Dict[str, Any], tols: Tolerances) ->
NightlyResult:
168         """Diff current run vs frozen baseline.
169
170         * Segmenter changed ? STALE (numbers not comparable; re-freeze).
171         * Corpus (video set) changed ? STALE + report added/removed; still diff the
```

```

INTERSECTION.
172         * Per shared video: reel_count EXACT (any change = regression); a pipeline ERROR =
regression;
173         quality metrics drop beyond ``quality_tol`` = regression (rise = improvement).
174         """
175         res = NightlyResult()
176         if baseline.get("segmenter") != current.get("segmenter"):
177             res.findings.append(Finding("", "segmenter", "stale",
178                                     detail=f"{baseline.get('segmenter')} ?
{current.get('segmenter')}"))
179         return res
180
181         b_pv, c_pv = baseline.get("per_video", {}), current.get("per_video", {})
182         res.added = sorted(set(c_pv) - set(b_pv))
183         res.removed = sorted(set(b_pv) - set(c_pv))
184         if res.added or res.removed:
185             res.findings.append(Finding("", "corpus", "stale",
186                                     detail=f"added={res.added} removed={res.removed}"))
187
188         for v in sorted(set(b_pv) & set(c_pv)):
189             b, c = b_pv[v], c_pv[v]
190             # 1) pipeline must still succeed
191             if not c.get("ok", False) or c.get("reel_count") is None:
192                 res.findings.append(Finding(v, "error", "regression",
193                                     detail=str(c.get("error") or "no reel_count
produced")))
194             continue
195             # 2) reel count ? EXACT
196             bc, cc = b.get("reel_count"), c.get("reel_count")
197             if bc is not None and cc is not None and cc != bc:
198                 res.findings.append(Finding(v, "reel_count", "regression", float(bc),
float(cc)))
199             # 3) quality metrics ? tolerance (only if both sides have them)
200             bq, cq = b.get("quality") or {}, c.get("quality") or {}
201             for m in QUALITY_HIGHER:
202                 bv, cv = bq.get(m), cq.get(m)
203                 if bv is None or cv is None:
204                     continue
205                 if cv < bv - tols.quality_tol:
206                     res.findings.append(Finding(v, m, "regression", bv, cv))
207                 elif cv > bv + tols.quality_tol:
208                     res.findings.append(Finding(v, m, "improvement", bv, cv))
209         return res
210
211

```

38. assistant (2026-06-22T09:43:29.327Z)

Now let me look at the critical issue with the cost calculation and the tolerance check logic more carefully:

39. user (2026-06-22T09:43:34.516Z)

```

cv=0.39, bv - tol=0.38, is_regression: False
cv=None, would skip: True
bv=None, cv=0.3, would skip: True

```

40. assistant (2026-06-22T09:43:36.436Z)

Let me check the critical edge case with the re-run guard and error handling:

41. user (2026-06-22T09:43:40.742Z)

Exit code 1

```

Traceback (most recent call last):
  File "<string>", line 27, in <module>
    print(f'Will this PASS or FAIL vs baseline={baseline_count}? {FAIL if count !=
baseline_count else PASS}')
                                                                    ^^^^^
NameError: name 'FAIL' is not defined
Re-running because count mismatch...
Adopted second run count: 5
Final count: 5

```

42. user (2026-06-22T09:43:45.472Z)

```

Re-running because count mismatch...
Adopted second run count: 5
Final count: 5
Will this PASS or FAIL vs baseline=7? FAIL

```

43. assistant (2026-06-22T09:43:47.402Z)

Now let me check the exit code mapping more carefully for correctness:

44. user (2026-06-22T09:43:49.178Z)

```

471     def main(argv: Optional[List[str]] = None) -> int:
472         import logging
473         logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")
474         log = logging.getLogger("nightly_reelcount")
475         args = build_parser().parse_args(argv)
476         tols = Tolerances(quality_tol=args.quality_tol)
477         date = args.date or _dt.date.today().isoformat()
478
479         if not os.path.exists(args.manifest):
480             log.error("manifest not found: %s", args.manifest)
481             return 2
482         manifest = load_manifest(args.manifest)
483         baseline = load_baseline(args.baseline)
484
485         try:
486             run_results, total_wall = run_corpus(manifest, baseline, rerun_on_mismatch=not
args.no_rerun_guard)
487             except Exception as e: # noqa: BLE001 ? operational failure (deps/config) ? exit 2,
never a false "pass"
488                 log.error("run failed: %s", e)
489                 return 2
490         if not run_results:
491             log.error("no videos in manifest %s", args.manifest)
492             return 2
493
494         billed = args.vm_uptime_seconds if args.vm_uptime_seconds is not None else
total_wall
495         cost = cost_report(
496             billed_seconds=billed,
497             machine_usd_per_hr=float(manifest.get("machine_usd_per_hr",
DEFAULT_MACHINE_USD_PER_HR)),
498             fx_inr_per_usd=float(manifest.get("fx_inr_per_usd", DEFAULT_FX_INR_PER_USD)),
499             gpu_seconds=total_wall,
500         )
501         segmenter = manifest.get("segmenter", "motion")
502         snapshot = summarize_results(run_results, segmenter, cost)
503         snapshot["git_commit"] = _git_head()
504         snapshot["generated_at"] = date
505
506         if args.update_baseline:
507             save_baseline(args.baseline, snapshot)

```

```

508         log.info("baseline frozen ? %s", args.baseline)
509         for r in run_results:
510             log.info("  %s: reels=%s ok=%s", r["name"], r.get("reel_count"),
r.get("ok"))
511         return 0
512
513     if baseline is None:
514         log.error("no baseline at %s ? freeze one with --update-baseline (on the
configured engine).", args.baseline)
515         return 3
516
517     result = compare(baseline, snapshot, tols)
518     report = format_report(baseline, snapshot, result, date, snapshot["git_commit"])
519
520     if not args.no_report:
521         os.makedirs(args.report_dir, exist_ok=True)
522         md_path = os.path.join(args.report_dir, f"{date}.md")
523         with open(md_path, "w", encoding="utf-8") as fh:
524             fh.write(report + "\n")
525         with open(os.path.join(args.report_dir, f"{date}.json"), "w", encoding="utf-8")
as fh:
526             json.dump({"date": date, "status": result.overall_status(), "exit_code":
result.exit_code(),
527                       "snapshot": snapshot, "regressions": [f.message() for f in
result.regressions]}),
528                       fh, indent=2, sort_keys=True)
529         log.info("report ? %s", md_path)
530
531     print(report)
532
533     if args.email:
534         try:
535             from scripts.nightly_email import send_report
536             subject = f"[KhelSutra nightly] {result.overall_status()} ? reel-count
engine regression {date}"
537             sent = send_report(subject, report, to=args.email_to)
538             log.info("email %s", "sent" if sent else "skipped (no creds ? report still
written)")
539         except Exception as e: # noqa: BLE001 ? a broken mailer must NOT mask the
regression verdict
540             log.error("email failed (report still written): %s", e)
541
542     code = result.exit_code()
543     if code == 1:
544         log.error("REEL-COUNT REGRESSION (%d) ? see %s", len(result.regressions),
args.report_dir)
545     elif code == 4:
546         log.warning("baseline STALE ? corpus/segmenter changed; re-freeze with --update-
baseline")
547     else:
548         log.info("nightly reel-count check PASSED (cost $%.4f / ?%.2f)", cost["usd"],
cost["inr"])
549     return code
550

```

45. assistant (2026-06-22T09:43:51.616Z)

Now let me look for one critical issue with the exception handler that could mask regressions:

46. user (2026-06-22T09:43:56.924Z)

Exit 2: no videos processed

All error paths checked: 2 (operational), 3 (no baseline), should be OK

47. assistant (2026-06-22T09:43:59.161Z)

Let me look at critical issue with comparing when baseline video reel_count is None:

48. user (2026-06-22T09:44:06.167Z)

No regression detection

Edge case is safe due to the error check on line 191

49. assistant (2026-06-22T09:44:09.695Z)

Now let me check the quality comparison logic for a subtle bug with the tolerance:

50. user (2026-06-22T09:44:14.875Z)

```
bv=0.0, cv=-0.001, tol=0.02
Regression check: cv < bv - tol = -0.001 < -0.02 = False
```

```
bv=0.01, cv=0.0, tol=0.02
Regression check: cv < bv - tol = 0.0 < -0.01 = False
```

Tolerance logic is correct

51. assistant (2026-06-22T09:44:17.000Z)

Now let me check for SMTP security issue:

52. user (2026-06-22T09:44:18.473Z)

```
84         import smtplib
85         msg = build_message(subject, body_md, from_addr=user, to_addr=to_addr)
86         with smtplib.SMTP(host, port, timeout=60) as s:
87             s.starttls()
88             s.login(user, password)
89             s.send_message(msg)
90         log.info("emailed report to %s", to_addr)
91         return True
92
```

53. assistant (2026-06-22T09:44:20.715Z)

Now I've completed my thorough analysis. Let me compile the findings:

54. user (2026-06-22T09:45:09.039Z)

Structured output provided successfully

55. assistant (2026-06-22T09:45:21.713Z)

Summary

I've completed a thorough correctness and security review of the nightly reel-count engine regression pipeline. Here are the key findings:

HIGH-SEVERITY DEFECTS (production risk):

- **SMTP TLS vulnerability**** (`scripts/nightly\_email.py:86-87`): No explicit SSL context is provided to `smtplib.SMTP()`. Use `ssl.create\_default\_context()` explicitly.
- **Re-run guard fragility**** (`scripts/nightly\_reelcount.py:362-366`): The re-run logic adopts the second-run count without re-validating against baseline. While the design is correct (transient flakes won't produce two different results), there's no visibility if the two runs

produce different counts. Add observability logging.

MEDIUM-SEVERITY DEFECTS (quality/robustness):

3. ****Quality metrics silently skip when baseline lacks labels****
(`scripts/nightly\_reelcount.py:200-208`): No issue functionally, but by design newly-measured quality metrics are never compared. Acceptable but document it.
4. ****Deep segmenter crashes mask as operational failures****
(`scripts/nightly\_reelcount.py:485-489`): Any exception in run_corpus() exits 2 (operational), not 1 (regression). This is intentional but could hide determinism bugs.

LOW-SEVERITY DEFECTS (optimization/hardening):

5. ****Unvalidated cost rate**** (`scripts/nightly\_reelcount.py:79`): `machine\_usd\_per\_hr` isn't validated; negative values produce negative costs.
6. ****Baseline check deferred until after GPU run**** (`scripts/nightly\_reelcount.py:513-515`): Wasted compute time (5-10 min) if baseline is missing?check earlier.
7. ****SMTP password error not distinguished**** (`scripts/nightly\_email.py:86-89`): Auth failures raised as exceptions; wrap in try-except to align with graceful fallback design.
8. ****GCP VM metadata not validated**** (`deploy/gcp/nightly\_regression/run\_on\_vm.sh:20-26`): Missing or empty keys could cause failures mid-run; add early validation.

All findings are documented with file:line references and actionable fixes. The core comparison logic (reel-count exact, quality tolerance, segmenter/corpus stale detection) is sound?no path found where a real regression exits 0 or is silently masked as pass.